The three fundamental discoveries that quantum mechanics has led to are these: granularity, indeterminacy, and the relational aspect of physical variables. Each one of these demolishes further the little that was left of our idea of time. Let's consider them one by one.

GRANULARITY

The time measured by a clock is "quantified," that is to say, it acquires only certain values and not others. It is as if time were granular rather than continuous.

Granularity is the most characteristic feature of quantum mechanics, which takes its name from this: "quanta" are elementary grains. A minimum scale exists for all phenomena.[51] For the gravitational field, this is called the "Planck scale." Minimum time is called "Planck time." Its value can be easily estimated by combining the constants that characterize phenomena subject to relativity, gravity, and quantum mechanics.[52] Together, these determine the time to 10^{-44} seconds: a hundred millionth of a trillionth of a trillionth of a trillionth of a second. This is Planck time: at this extremely minuscule level, quantum effects on time become manifest.

Planck's time is small, much smaller than any actual clock is today capable of measuring. It is so extremely small that we should not be astounded to discover that "down there," at such a minute scale, the notion of time is no longer valid. Why *should* it be? Nothing is valid always and everywhere. Sooner or later, we always come across something that is new.

The "quantization" of time implies that almost all values of time *t do not exist*. If we could measure the duration of an interval with the most precise clock imaginable, we should find that the time measured takes only certain discrete, special values. It is not possible to think of duration as continuous. We must think of it as discontinuous: not as something that flows uniformly but as something that in a certain sense jumps, kangaroo-like, from one value to another.

In other words, a *minimum* interval of time exists. Below this, the notion of time does not exist—even in its most basic meaning.

Perhaps the rivers of ink that have been expended discussing the nature of the "continuous" over the centuries, from Aristotle to Heidegger, have been wasted. Continuity is only a mathematical technique for approximating very finely grained things. The world is subtly discrete, not continuous. The good Lord has not drawn the world with continuous lines: with a light hand, he has sketched it in dots, like the painter Georges Seurat.